

SnowDay Report

An Agent Submitted to ANAC 2026 Human-Agent Negotiation League

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Abstract

We present **SnowDay**, our submitted agent for the ANAC 2026 Human-Agent Negotiation League. While Large Language Models (LLMs) are useful communicators, they are unreliable strategic decision-makers without additional structure. Our agent makes all decisions based on a lightweight symbolic negotiation policy, using an online classifier to infer the opponent’s behavior style from early offers and select a corresponding offer strategy for our agent.

To evaluate SnowDay beyond the official leaderboard score, we introduce **NegoBench**, a diagnostic evaluation framework for the HAN setting. Using this framework to evaluate SnowDay, we find that classifier-guided per-type offer strategy improves over a fixed-curve baseline. In its strongest HAN tournament run, SnowDay ranked 1st out of 20 agents with normalized advantage approximately 0.60; in local LLM-persona evaluations, it reached 0.618 mean advantage. The ablations we conduct show that type-conditioned decision-making improves both mean advantage and agreement rate, making us competitive against varied HAN opponents.

1 Introduction

The HAN league pits agents against a pool of personality-adjusted LLM negotiators in a qualifying round and against real humans in the finals, under the Stacked Alternating Offers protocol with linear additive private utilities, private reservation values r , and a $\sim 300s/\sim 40$ -round time budget. The scoring metric is normalized advantage, which is given by:

$$A = \frac{U(\omega^*) - r}{\max_{\omega} U(\omega) - r} \in [0, 1],$$

with a score of zero on disagreement. A deal barely above the reservation value is worth almost nothing, so the central challenge of the game is for an agent to attain as much utility as possible while still reaching agreement against varied opponents.

We present **SnowDay**¹, our submitted HAN agent. It follows the BOA architecture of negotiation agents, decomposed into a bidding strategy, an opponent model, and an acceptance condition [1]. The main approach of SnowDay is an augmentation to the Opponent modeling module; specifically, we add a learned opponent-type classifier that maps early offer behavior to a behavioral type, which then selects a type-conditioned offer

strategy. We hypothesize (and demonstrate in later sections) that this type-based classifier is reasonably effective in the HAN setting: because the opponent pool consists of distinct personality-adjusted negotiators, identifying these behaviors early on allows SnowDay to develop a behavior-aware strategy.

Alongside our agent, we also introduce **NegoBench**, a custom evaluation framework, which we believe is important for two main reasons. First, the feedback from official HAN runs are slow and unstable, making it an unreliable signal for ablations on SnowDay’s design. Second, more importantly, we believe that the normalized advantage A is insufficient to fully evaluate an automated negotiator. For instance, it fails to capture whether an agent struggles against specific opponent types (e.g. hardliners vs conceders), is susceptible to manipulation through language, or misses higher-utility offers in earlier parts of negotiation. NegoBench provides us more details into these gaps by decomposing negotiation capability into measures of numerical utility, opponent modeling, and communication sensitivity. We explain these further in Section 4.

2 Related Work

Automated Negotiation and the BOA Architecture. SnowDay builds on the BOA architecture, which decomposes a negotiator into Bidding strategy, Opponent model, and Acceptance condition components [1]. SnowDay is a HAN-specific instantiation of BOA: its bidding/offer strategy is a type-conditioned concession curve, its opponent model is an online behavioral classifier and an opponent preference estimator, and its acceptance condition is tuned to maximize its own utility while avoiding no-deals (i.e. timed out negotiations or opponent walk outs).

Opponent Modeling: Preference Modeling and Behavior Type Classification. BOA-style agents often use opponent models to estimate the opponent’s preferences from bidding history. Frequency-based models, for example, infer likely issue priorities from how often issue values appear in the opponent’s offers [1, 6].

On top of this, our main addition is behavioral opponent modeling. Using the opponent’s early offer behavior, SnowDay classifies the opponent into a behavioral type, such as conceder, fairish, hardliner, or wallish. This predicted type determines which offer strategy to use, represented by different concession curves. We discuss this further in Section 3.

Negotiation Benchmarks. Recent work has benchmarked LLMs in negotiation settings. NegotiationArena [3] evaluates

¹The name evolved from an earlier agent, SunnyDay, named after one of our teammates, whose name evolved into RainyDay, when performance sunk, but was eventually elevated to SnowDay.

LLM agents in open-ended scenarios like resource allocation and price negotiation, reporting outcomes such as payoffs, agreement rates, and behavioral tactics. NegotiationToM [4] focuses on whether models can infer opponents’ beliefs, desires, and intentions. Our NegoBench evaluation framework draws inspiration from these benchmarks, treating negotiation performance as more than a final payoff like NegotiationArena [3], and measuring opponent understanding like NegotiationToM [4]. However, NegoBench adapts these ideas to the narrower and more controlled HAN setting of finite outcome spaces, additive private utilities, and a fixed pool of personality-adjusted opponents.

3 SnowDay Agent Design

The main design of our agent follows the BOA architecture, as illustrated in Figure 1. For language, we use an LLM-free design, using deterministic sentences for the offer’s text field. We break down the design into the Opponent modeling, Bidding/offer strategy, and Acceptance condition in the following subsections.

3.1 Opponent Type Classifier

The core of the approach is using an online classifier that selects a per-type offer strategy. The type classifier decides how aggressively SnowDay should bargain (i.e. the offer curve), while the frequency preference model helps choose which partner-compatible offer to make that still preserves our utility. This is illustrated in the topmost blue box of Figure 1.

Features. From the opponent’s first $K = 4$ offers we extract the following features:

Feature	Meaning
n_offers_seen	Number of valid opponent offers observed, divided by K
opening_advantage	Normalized advantage of the opponent’s first offer
latest_advantage	Normalized advantage from their latest observed offer
best_advantage	Best normalized advantage they have offered us so far
mean_advantage	Average normalized advantage of their observed offers
concession_slope	Linear trend of their offers from our perspective. Positive = offers are improving for us over time Difference between latest and opening offer. Positive means they conceded overall.
total_concession	Average issue-value distance between consecutive offers. High means they change offers a lot; Low means they repeat similar offers.
mean_offer_movement	Fraction of repeated opponent offers
repeat_rate	Fraction of issues where the same value appears in $\geq 80\%$ of observed offers
stubborn_issue_fraction	No. of opponent turns since they last improved their offer for us, divided by K

Table 1: Opponent Type MLP-based classifier features.

Model and labels. A small Multilayer Perceptron (MLP) with $11 \rightarrow 64 \rightarrow 32 \rightarrow 4$ perceptrons at each layer and ReLU activation function maps the prefix vector to a type in:

1. *Conceder*: Improves its offers over time, moving toward deals that are better for us.
2. *Fairish*: Tends toward balanced or mutually reasonable deals.
3. *Hardliner*: May concede late or only a little. In terms of MLP’s features, it may show high `stubborn_issue_fraction`.

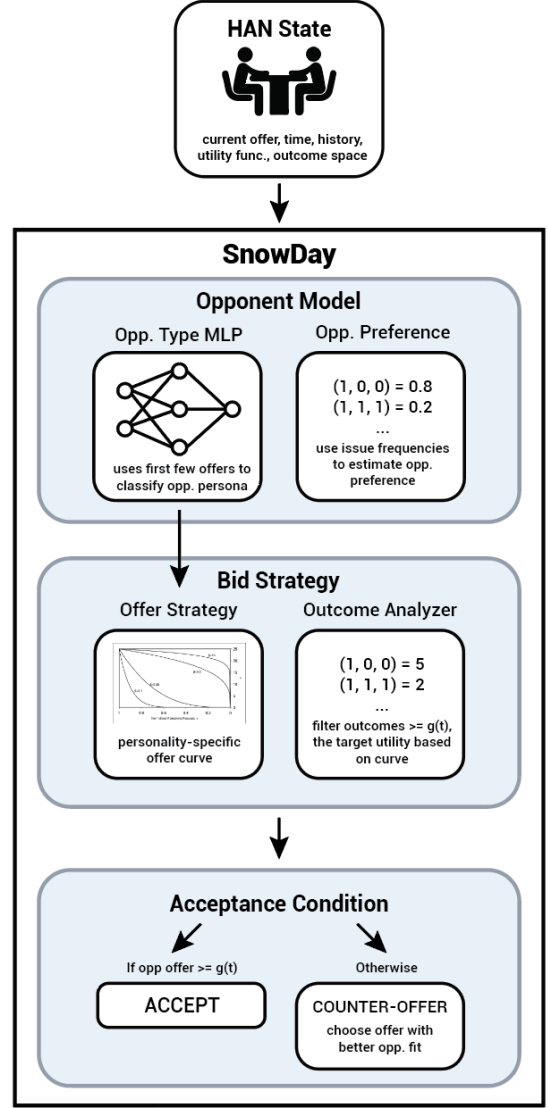


Figure 1: SnowDay decision pipeline. The agent tracks negotiation history, extracts early-offer features, predicts an opponent type with an MLP classifier, selects a type-conditioned concession curve, and chooses whether to accept or counteroffer.

4. *Wallish*: An opponent that behaves like a wall, such as having repeated selfish offers, or, in terms of MLP’s features, having high `rounds_since_improvement` or `repeat_rate`.

At inference, the agent re-classifies online as offers accumulate and updates its active curve. The trained network is then exported to a JSON file.

3.2 Frequency-Based Opponent Preference Model

The opponent type classifier determines the offer strategy curve, but it does not by itself determine which concrete outcome to offer. To do this, SnowDay uses a frequency-based opponent preference model. For each issue, SnowDay counts how often the opponent has proposed each value. A candidate offer receives

Opponent Type	f	E	Response
Conceder	0.45	1.0	wait for opponent concessions; avoid early surplus loss
Fairish	0.55	1.0	meet in the middle
Hardliner	0.35	2.2	hold longer, then drop at the end
Wallish	0.05	1.0	salvage above-reserve deal near deadline

Table 2: The f and E values we used in the Boulware curve for SnowDay agent, derived via grid search.

one score per issue:

$$\text{fit}_i(\omega) = \frac{\text{count of the candidate's value on issue } i}{\text{count of the opp's most frequent value on issue } i}.$$

The total opponent-fit score is the average across issues:

$$\text{Fit}(\omega) = \frac{1}{n} \sum_{i=1}^n \text{fit}_i(\omega).$$

A score of 1 means the candidate offer uses the opponent’s most frequently proposed value on every issue. A score near 0 means it uses values the opponent proposed only a few times. This gives SnowDay a lightweight estimate of which offers are likely to look acceptable to the opponent, without knowing the opponent’s true utility function. Ties between offers are broken by SnowDay’s own normalized advantage.

3.3 Type-Conditioned Bidding/Offer Strategy

After classifying an opponent type, SnowDay follows a corresponding Boulware curve [5] for its offer with respect to the normalized time t of the negotiation

$$g(t) = f + (s - f) (1 - t^E),$$

with start fraction $s = 1$, final fraction f , and concession exponent E (Table 2). Intuitively, f dictates the absolute floor of our demands (i.e. how much we concede overall), and E dictates the rate at which we concede (e.g. whether we drop our target linearly or hold out stubbornly until the deadline). Among outcomes whose normalized utility is at least $g(t)$, SnowDay selects the outcome with the highest partner-fit score under its frequency-based preference model. When remaining time for a negotiation runs low, we lower the target utility toward a hand-coded floor, preferring to reach an agreement with a smaller surplus over our reserve rather than a timeout no-deal.

To find the curve for each type, we sweep a discrete grid of curve parameters (s, f, E, rescue) and keep the offer strategy that gave our agent the highest mean normalized advantage minus penalties for zero-score outcomes and timeouts against the deterministic opponents we use in NegoBench. While the current approach is hyperparameter search using grid values, we can interpret the strategy shown in Table 2: for example, for a predicted fair opponent, SnowDay uses a slightly higher final target, $f = 0.55$, with a linear concession schedule. This aligns with the expectation that a fair opponent will meet us in a reasonable middle region, so the agent should avoid dropping too low while still making visible progress toward agreement. We illustrate the derived curves in Figure 2.

3.4 Acceptance Condition

At each turn, the agent first computes the counteroffer it would make if it rejected the current offer. It accepts the opponent’s

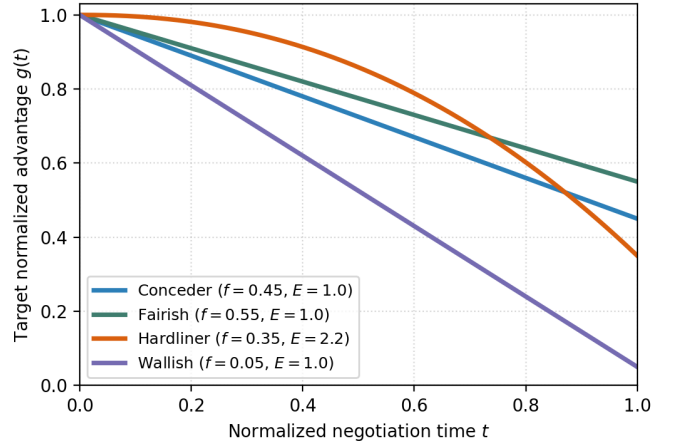


Figure 2: Offer curves for the different opponent types.

current offer when that offer gives at least as much normalized advantage as the planned counteroffer, provided that the offer is above the agent’s reservation value. Near the deadline (i.e. when $t \geq 0.97$), the agent accepts any above-reservation offer.

3.5 Text Communication

After the bidding strategy selects an offer (or chooses to accept), SnowDay produces a concise message from pre-determined templates we created. The template is conditioned on (1) the inferred opponent behavioral type $\in \{\text{conceder, fairish, hardliner, wallish}\}$ and (2) the current negotiation phase (i.e. the opening of the negotiation, the point in time when SnowDay has identified an opponent type, and the point in time near the deadline $t \geq 0.97$). This directs the tone and emphasis without changing the underlying offer or counter-offer.

Considering the human aspect to HAN, to ensure that the language matches the actual offer being sent, the text is constructed directly from the same outcome tuple used in the SAO response. Thus, if the formal offer changes—for example, the quantity of apples or oranges in the Grocery scenario, the text refers to those same issue-value assignments.

4 NegoBench Evaluation Framework

As mentioned, the HAN leaderboard scoring metric is normalized advantage:

$$A_i(\omega) = \frac{U_i(\omega) - r_i}{\max_{\omega'} U_i(\omega') - r_i},$$

with zero score on disagreement. However, this compresses different negotiation abilities into one number. To provide more detail than just a single scalar score, NegoBench decomposes an agent’s performance on five diagnostic pillars: value claiming, value creation, opponent modeling, temporal/acceptance behavior, and communication sensitivity. These pillars are primarily measured by pitting SnowDay against a fixed pool of LLM-free, **deterministic opponents**: hardcoded negotiators with known utility functions, known acceptance rules, and controlled personality-like behaviors such as concessionary, fair, hardline, misleading, revealing, and wall-like bargaining. We

also report empirical normalized advantage and agreement rates against stochastic LLM-powered negotiators.

Pillar 1: Value claiming. This measures how much utility the agent attains for itself when agreement is possible. The HAN normalized advantage is the primary metric. However, raw advantage is not always comparable across opponent types: against a hardliner opponent, very high advantage may be unreachable without causing disagreement, while against a conceder, the same score may leave value on the table. For **deterministic opponents** (i.e. opponents with a hardcoded acceptance strategy), NegoBench can compute a reachable ceiling

$$A_i^* = \max_{\omega \in \mathcal{W}} A_i(\omega),$$

where $\mathcal{W}$ is the set of outcomes that some deterministic opponent would accept under its fixed acceptance rule. This allows us to report reachable **oracle efficiency**, which is given by

$$\text{Eff}_i = A_i / A_i^*.$$

Again, note that this oracle ceiling is only used for deterministic opponent policies. For stochastic or LLM-driven opponents, NegoBench reports normalized advantage and agreement rate, as we cannot claim an exact reachable ceiling.

Pillar 2: Value creation. This pillar asks whether the deal reaches an outcome that is more efficient for both parties together. Because NegoBench has access to both agents' utility functions at evaluation time, we can enumerate every possible outcome/agreement and check whether the final one lies on the Pareto frontier. We use the frontier rate, which is the share of agreements/outcomes that land exactly on the Pareto frontier, as our main metric. We also report the Pareto distance: the Euclidean distance, in min-max normalized utility space (i.e. worst outcome for one player's side is 0 and the best outcome is 1), from the agreement to the nearest Pareto-frontier point.

Pillar 3: Opponent Modeling. This measures how well the agent infers opponent's behavioral type. In our deterministic opponent pool, each rollout has a known ground-truth opponent type, so we can directly report type-classification accuracy.

We also measure whether opponent type classification improves negotiation outcomes. To do so, we evaluate the normalized advantage scores obtained by three policies on the same fixed opponent pool: a fixed-curve baseline with no type information, the deployed classifier-conditioned policy, and an oracle policy that selects the curve using the true type label. The difference between the classifier-conditioned policy and the fixed baseline measures the value of type-conditioned bidding. The difference between the oracle policy and the classifier-conditioned policy measures the cost of classification error. Mathematically, these are given by:

$$\Delta_{\text{type}} = \bar{A}(\pi_{\text{classifier}}) - \bar{A}(\pi_{\text{fixed}}),$$

$$\Delta_{\text{error}} = \bar{A}(\pi_{\text{oracle}}) - \bar{A}(\pi_{\text{classifier}}),$$

where $\bar{A}(\pi)$ is mean normalized advantage on a fixed opponent pool achieved by policy π .

Pillar 4: Temporal and acceptance behavior. This measures whether an agent manages time well under the alternating-offers protocol. A negotiator can fail even with good candidate offers if it concedes too early, holds out too long, or rejects offers that it should accept before the deadline. We evaluate both agreement rate and stopping regret. Stopping regret is the gap between the best incoming offer the agent could have accepted in hindsight and the value it achieved in the end. This allows us to identify if an agent rejected a good offer and later timed out or settled for less.

Pillar 5: Communication Sensitivity. To isolate the impact of opponent language's effect to our agent, NegoBench uses behavior-matched *twin* opponents. Here, the twins make the same issue offers and use the same acceptance rule, but only differ in their messages. One twin is silent, one truthfully reveals priorities, and one gives misleading information about what issues they prioritize. Because the offer strategy of our opponent is fixed, differences in agent performance can be attributed to the language channel.

Let $\bar{A}(\pi, o)$ denote the score (normalized advantage) of agent policy π against opponent o . We define the truth premium as

$$\text{TruthPremium} = \bar{A}(\pi, o_{\text{truth}}) - \bar{A}(\pi, o_{\text{silent}}),$$

and the deception cost as

$$\text{DeceptionCost} = \bar{A}(\pi, o_{\text{silent}}) - \bar{A}(\pi, o_{\text{mislead}}).$$

A positive truth premium means that the agent benefits from honest preference disclosure, and a positive deception cost means that the agent is susceptible to misleading language. For SnowDay, which does not use language to make strategic decisions, we expect little or no change in its decisions across the twins. Future agents that use LLM-generated persuasion can be evaluated on whether language improves agreement quality without relying on deception.

Opponent Pools. NegoBench evaluates our agent against several pools of opponents: non-LLM agents, deterministic opponent agents, and personality-adjusted LLM-driven negotiators.

5 Experiments

Having defined the NegoBench pillars, we now use them to evaluate SnowDay. We first report the raw utility SnowDay attains on our local evaluation suite and on the official HAN runs. We then use NegoBench to give us insight on why it achieves these results by breaking its performance down across the five pillars.

5.1 Normalized Advantage Scores

On the local pool of deterministic opponents (108 matches), SnowDay reaches mean normalized advantage 0.503 with 100% agreement (108/108) and no failed negotiations.

On the real-LLM pool (72 matches), which more closely resembles the actual qualifying round, SnowDay's mean normalized advantage rises to 0.618 at 86% agreement (10 no-deals out of 72). It is strong against conceding, linear, and instant

Setting	Mean	Q1	Med	Speed
No-LLM personas (local)	.503	.400	.520	4 ms
LLM personas (local)	.618	.410	.690	4 ms
Tournament Run 19098	.60	.395	.65	4 ms

Table 3: SnowDay’s normalized advantage across evaluation settings. The first two rows are local evaluations: deterministic non-LLM personas and the six built-in personality-adjusted LLM negotiators from `negmas_llm` (Boulware, Conceder, Linear, Nice, Tough, Hybrid). The final row reports the official HAN tournament 19098 result. The Speed column denotes SnowDay’s response time.

Pillar	Metric Description	Numerical Score
1	Reachable Advantage Attained	0.884
2	Pareto Frontier Rate	0.750
3	Type Classification Accuracy	0.746
4	Agreement Rate	1.000
5	Truth Premium / Deception Cost	0.000 / 0.000

Table 4: NegoBench per-pillar results for SnowDay. Notably, SnowDay is able to achieve reasonably high advantage, classification accuracy on simulated runs, and reaches a 100% agreement rate against our deterministic opponents.

opponents; its main weakness is a tough LLM.Tough opponent, which accounts for 8 of those 10 no-deals.

Beyond our local evaluation suite, we report SnowDay’s strongest performance on an official tournament run (ID 19098: 20 agents, 5 configurations, 480 negotiations, 06/11/2026, 05:06:12 PM). In this tournament, SnowDay ranked 1st out of 20 agents, with a normalized advantage of approximately 0.60 and a median of 0.65..

5.2 NegoBench Pillar Analysis

NegoBench explains where SnowDay’s performance comes from. Table 4 gives its full per-pillar performance, measured on **deterministic opponent** pools. This distinction matters because oracle-based quantities such as reachable efficiency and Pareto-frontier diagnostics require access to the opponent’s policy or utility function; for stochastic LLM opponents, we report only empirical normalized advantage and agreement rate.

Pillar 1: Value claiming. SnowDay still secures high returns against tough opponents. Averaged over the two hardest deterministic types, hardliner and wall, it attains 0.884 of the reachable advantage. What this means is that even against opponents unwilling to concede, it claims most of the value that was available.

Pillar 2: Value creation. SnowDay’s agreements land on the Pareto frontier 75% of the time, showing that it does not only care about its own share but also rarely wastes value the two sides could have shared. A mean Pareto distance of just 0.014 further indicates that, once it closes a deal, the outcome is usually close to efficient.

Policy	Mean off. adv	Agree	No-deal	Online acc.
No-Classifier	.461	89%	11%	—
Classifier	.503	100%	0%	.722
Oracle	.517	100%	0%	1.00

Table 5: Ablation of opponent type conditioning on deterministic opponent personas. The classifier policy improves mean advantage and reduces no-deals relative to the no-classifier baseline, while the oracle policy estimates the upper bound from perfect type information.

True Label \ Prediction	Con.	Fair	Hard	Wallish	Accuracy
Conceder	59	36	1	0	.61
Fairish	8	142	1	33	.77
Hardliner	0	1	31	50	.38
Wallish	0	0	0	214	1.00

Table 6: Opponent-type confusion on the held-out test split (counts, $n=576$).

Pillar 3: Opponent modeling. This pillar evaluates whether SnowDay can identify the opponent’s behavioral type, and if using that predicted type improves negotiation outcomes.

Accuracy of MLP Classifier. To train and evaluate our opponent classifier, we collected a dataset of 2880 trajectories, which we split into training and held-out test sets. The classifier achieves 0.774 accuracy on the held-out test split and 0.746 accuracy on the full trajectory dataset. We notice that many errors occur between similar behaviors, such as conceder being classified as fairish, or hardliner being classified as wallish, as shown in Table 6. Although hardliner label accuracy is low, most hardliners are predicted as wallish, which selects a similarly conservative curve.

Ablation on Classifier. To test whether type information actually helps, we compare three offer strategies on the same deterministic opponent pool (Table 5). The *no-classifier baseline* ignores opponent type and uses the same offer strategy against every opponent. The deployed *classifier* policy selects its offer strategy based on the predicted opponent type, although this prediction is sometimes wrong. The *oracle* policy (i.e. the ground-truth) selects the offer strategy using the opponent’s true type. This is not a deployable policy, since the true opponent type is not known in real matches, but it serves as an upper bound on what a perfect classifier could achieve.

Empirically, the baseline reaches a mean advantage of 0.461 with an 11% no-deal rate (the opposite of agreement rate). The deployed classifier improves the mean advantage to 0.503 and reduces the no-deal rate to 0%. The oracle reaches 0.517. Therefore, the full value of perfect type information is the baseline-to-oracle gap, 0.056, while our deployed classifier captures 0.042 of that gain, or about 75%.

Pillar 4: Temporal and Acceptance Behavior. SnowDay reaches agreement on every match in the deterministic pool. This shows that its time management and near-deadline acceptance effectively avoid no-deals without sacrificing much utility by conceding too early.

Pillar 5: Communication Sensitivity. SnowDay’s decisions do not depend on language. Its behavior is therefore identical whether the opponent is silent, truthful, or misleading. This opens to future work on evaluating agents that communicate via natural language and interpret opponent language (i.e. LLM-based agents).

6 Conclusion and Future Work

We present **SnowDay**, our submitted agent for the ANAC 2026 Human-Agent Negotiation League, and **NegoBench**, the evaluation framework we built to understand its behavior. SnowDay places every strategic decision inside a lightweight symbolic policy: an online classifier reads the opponent’s first few offers, predicts its behavioral type, and selects a type-conditioned offer strategy accordingly. Our ablations show that selecting the offer strategy by predicted opponent type improves both advantage and agreement rate. Beyond this single leaderboard score, NegoBench decomposes performance into five pillars, allowing us to identify where the advantage comes from in a faster and more stable way than relying on the sparse HAN tournament feedback.

Given the time and resources available in completing this project, there are several limitations and significant room for future work.

- **Learning the curves through training.** SnowDay’s per-type curves are grid-searched. A natural next step is to replace these swept curves with a learned policy, following reinforcement-learning approaches such as RLBOA [2]. More generally, future work could learn the opponent model and bidding policy end-to-end. The agent could maintain a continuous latent representation of the opponent and then condition on this latent opponent representation when choosing whether to accept, counteroffer, etc.
- **Improving NegoBench for non-deterministic opponents.** Currently, some NegoBench diagnostics, especially reachable oracle efficiency, rely on deterministic opponent policies. This can only be reliably measured against hardcoded, deterministic opponents. Future work can explore how these metrics can be measured against stochastic or LLM-driven opponent pools.
- **Modeling more opponents.** We learned four opponent personalities. As discussed in the previous paragraph, we could instead maintain a continuous latent representation of the opponent without needing to bucket or discretize the opponent model.
- **Learning to talk.** Every prior RL negotiator learns only what to offer; HAN is LLM-mediated and our twins pool can causally isolate language, so a language-direction action with a twins-style reward is an open contribution.

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